

DDD and Event Studies

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bookdown::render_book()

Overview

Topics:

1. Difference-in-Difference-in-Differences (DDD)
2. Event Studies

Chapter 1

Difference-in-Difference-in-Differences

We'll bring back Cunningham and Cornwell (2013) to assess the impact of abortion on long-term gonorrhea incidences. The abortion hypothesis from Gruber, Levine, and Staiger (1999) makes very specific predictions about reproductive health and poverty outcomes. If there are far-reaching effects of abortion legalization in the early 1970s, then some of those effects will show up later. Levitt (2004) finds controversial evidence that abortion legalization caused a 10% decline in crime between 1991 and 2001.

The authors' focus on long-term gonorrhea cases is due to the correlation between single-parent status and risky behaviors. The authors focus on 15-19 year olds, since they would have been treated in the 5 early adopter states before complete legalization.

The authors use 25-29 year olds for a comparison group. The reason is that 25-29 year olds would be exposed to treatment but should not be affected by the treatment but close enough to 15-19 to prevent spillover effects.

1.1 The Estimator

Our Difference-in-Difference-in-Differences (DDD) is the following:

$$y_{ijt} = \alpha + \psi X_{ijt} + \beta_1 \tau_t + \beta_2 \delta_j + \beta_3 D_i + \beta_4 (\delta * \tau)_{jt} + \beta_5 (\tau * D)_{it} + \beta_6 (\delta * D)_{it} + \beta_7 (\gamma * \tau * D)_{ijt} + \varepsilon_{ijt}$$

All interactions are included in a DDD design

1. Treatment group dummy D_i
2. Post-treatment dummy τ_t
3. Group dummy δ_j
4. X_{ijt} is a set of covariates of interest

Where

1. D_i represents treatment state (5 adapters) and our control states
2. τ_t represents prepost-post period 1986-1992 and post-1992
3. δ_j represents our two groups of 15-19 vs 25-29 year olds

Our parameter of interest is β_7 to compare our DDD estimate to our original DD estimate. β_7 is the full-interaction term. We want to test β_7 to see if it is similar to our original difference-in-difference estimate.

Our diff-in-diff parameter for the placebo group is β_5 . We should expect to fail to reject the null hypothesis that $H_0 : \beta_5 = 0$.

1.2 15-19 year olds vs. 20-24 year olds

Let's read in our data and prepare it. We'll set up our demographic groups first.

```
cd "/Users/Sam/Desktop/Econ 672/Data/"
use "abortion.dta", clear

gen yr=(repeal) & (younger==1)
*White Male
gen wm=(wht==1) & (male==1)
*White Female
gen wf=(wht==1) & (male==0)
*Black Male
gen bm=(wht==0) & (male==1)
*Black Female
gen bf=(wht==0) & (male==0)

char year[omit] 1985
char repeal[omit] 0
char younger[omit] 0
char fip[omit] 1
char fa[omit] 0
char yr[omit] 0
```

```
/Users/Sam/Desktop/Econ 672/Data
```


Then we will estimate the DDD estimate for 15-19 year olds vs. 20-24 year olds in repeal vs Roe states without fixed effects.

```
reg lnr i.repeal##i.year##i.younger i.fip##c.t acc pi ir alcohol crack poverty income ur if bf==
```

```
(sum of wgt is 83,359,689)
note: 53.fip omitted because of collinearity.
note: t omitted because of collinearity.
note: 53.fip#c.t omitted because of collinearity.
```

```
Linear regression                               Number of obs   =       1,437
                                                F(41, 50)       =           .
                                                Prob > F        =           .
                                                R-squared       =       0.9152
                                                Root MSE       =       .25462
```

(Std. err. adjusted for 51 clusters in fip)

	lnr	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.repeal		.1632272	.2576836	0.63	0.529	-.3543455	.6807999
year							
1986		-.165378	.1279892	-1.29	0.202	-.4224519	.0916958
1987		-.3698162	.1391357	-2.66	0.011	-.6492785	-.0903539
1988		-.3800703	.1566106	-2.43	0.019	-.6946319	-.0655087
1989		-.3842067	.1766438	-2.18	0.034	-.7390063	-.0294071
1990		-.4217426	.1995766	-2.11	0.040	-.8226039	-.0208813
1991		-.5451351	.2281037	-2.39	0.021	-1.003295	-.0869754
1992		-.7745616	.2802143	-2.76	0.008	-1.337389	-.2117346
1993		-1.251466	.4434538	-2.82	0.007	-2.142169	-.3607631
1994		-1.07897	.3129618	-3.45	0.001	-1.707572	-.4503677
1995		-1.400012	.3944501	-3.55	0.001	-2.192288	-.6077355
1996		-1.498323	.3861419	-3.88	0.000	-2.273911	-.7227338
1997		-1.542576	.4198254	-3.67	0.001	-2.38582	-.6993321
1998		-1.471983	.4647098	-3.17	0.003	-2.40538	-.5385856
1999		-1.516729	.4854222	-3.12	0.003	-2.491728	-.5417294
2000		-1.556858	.5588659	-2.79	0.008	-2.679373	-.4343426
repeal#year							
1 1986		.1532595	.1528445	1.00	0.321	-.1537378	.4602568
1 1987		.0897675	.1613556	0.56	0.580	-.2343248	.4138598
1 1988		-.2081356	.2326221	-0.89	0.375	-.6753708	.2590997
1 1989		-.5113889	.2844963	-1.80	0.078	-1.082817	.0600388

1	1990		-.6599403	.2074843	-3.18	0.003	-1.076685	-.2431959
1	1991		-.8084183	.205713	-3.93	0.000	-1.221605	-.3952316
1	1992		-.7751547	.2643143	-2.93	0.005	-1.306046	-.2442638
1	1993		-.4947027	.3389076	-1.46	0.151	-1.175419	.1860132
1	1994		-.8088745	.2374756	-3.41	0.001	-1.285858	-.3318908
1	1995		-.6549101	.270693	-2.42	0.019	-1.198613	-.1112073
1	1996		-.9601593	.2358612	-4.07	0.000	-1.4339	-.4864181
1	1997		-1.120558	.2610365	-4.29	0.000	-1.644865	-.5962507
1	1998		-1.079713	.3078719	-3.51	0.001	-1.698092	-.4613344
1	1999		-1.135914	.3387383	-3.35	0.002	-1.81629	-.455538
1	2000		-1.18652	.4198777	-2.83	0.007	-2.02987	-.3431712
1.younger			.6982536	.1147531	6.08	0.000	.4677653	.9287419
repeal#younger								
1 1			-.0721284	.1258185	-0.57	0.569	-.3248423	.1805856
year#younger								
1986 1			.1165885	.1145906	1.02	0.314	-.1135735	.3467506
1987 1			.0876298	.1011899	0.87	0.391	-.115616	.2908757
1988 1			.0855004	.1289568	0.66	0.510	-.1735168	.3445177
1989 1			.088618	.1299416	0.68	0.498	-.1723774	.3496134
1990 1			.1158344	.1461543	0.79	0.432	-.1777252	.409394
1991 1			.1872056	.1276435	1.47	0.149	-.0691739	.4435851
1992 1			.2187408	.1389343	1.57	0.122	-.0603169	.4977984
1993 1			.5224545	.3594503	1.45	0.152	-.1995227	1.244432
1994 1			.2955902	.1126398	2.62	0.011	.0693465	.5218339
1995 1			.3851888	.1375715	2.80	0.007	.1088683	.6615093
1996 1			.3720165	.114905	3.24	0.002	.1412231	.6028099
1997 1			.3759459	.1164935	3.23	0.002	.1419618	.6099301
1998 1			.3535247	.1081798	3.27	0.002	.1362391	.5708102
1999 1			.3643688	.1195171	3.05	0.004	.1243115	.604426
2000 1			.3537662	.1166262	3.03	0.004	.1195157	.5880168
repeal#year#								
younger								
1 1986 1			-.3374017	.1148788	-2.94	0.005	-.5681426	-.1066607
1 1987 1			-.3893566	.1548238	-2.51	0.015	-.7003293	-.0783839
1 1988 1			-.3816576	.1428039	-2.67	0.010	-.6684877	-.0948276
1 1989 1			-.2773647	.1381061	-2.01	0.050	-.5547589	.0000295
1 1990 1			-.0460118	.1463254	-0.31	0.754	-.339915	.2478915
1 1991 1			.0790799	.147817	0.53	0.595	-.2178193	.3759791
1 1992 1			.1217134	.1401274	0.87	0.389	-.1597407	.4031675
1 1993 1			-.1680948	.3596801	-0.47	0.642	-.8905337	.554344
1 1994 1			.239085	.1235894	1.93	0.059	-.0091516	.4873217
1 1995 1			.1509293	.1416239	1.07	0.292	-.1335306	.4353892

1	1996	1		.1833938	.1143968	1.60	0.115	-.046379	.4131666
1	1997	1		.35702	.1140921	3.13	0.003	.1278592	.5861808
1	1998	1		.0960569	.1051921	0.91	0.366	-.1152277	.3073415
1	1999	1		.0967503	.1222801	0.79	0.433	-.1488565	.342357
1	2000	1		.0839178	.1334809	0.63	0.532	-.1841866	.3520221
	fip								
2				-.8120592	.3355711	-2.42	0.019	-1.486074	-.1380449
4				.1228053	.3134889	0.39	0.697	-.5068556	.7524662
5				-.1766227	.0298097	-5.93	0.000	-.2364972	-.1167482
6				-.1200182	.1449951	-0.83	0.412	-.4112493	.171213
8				-.0783986	.3136868	-0.25	0.804	-.708457	.5516599
9				-.6579097	.4593995	-1.43	0.158	-1.580641	.2648213
10				.0367743	.3593092	0.10	0.919	-.6849195	.758468
11				.1081184	.5422609	0.20	0.843	-.9810446	1.197281
12				.2535634	.3111603	0.81	0.419	-.3714205	.8785473
13				-.0944965	.1693867	-0.56	0.579	-.4347197	.2457267
15				-2.344423	.1920494	-12.21	0.000	-2.730165	-1.95868
16				-.6301702	.2210445	-2.85	0.006	-1.074151	-.1861892
17				-.3120264	.267051	-1.17	0.248	-.8484142	.2243614
18				.0038511	.2238371	0.02	0.986	-.445739	.4534413
19				-.0518566	.2232802	-0.23	0.817	-.5003282	.396615
20				.0487544	.2461016	0.20	0.844	-.4455552	.5430641
21				-.0881486	.058344	-1.51	0.137	-.205336	.0290388
22				-.8473393	.1650287	-5.13	0.000	-1.178809	-.5158693
23				-1.671066	.3661742	-4.56	0.000	-2.406548	-.9355832
24				-1.053722	.3446372	-3.06	0.004	-1.745947	-.3614982
25				-.5968925	.3391608	-1.76	0.085	-1.278117	.084332
26				.075925	.2916544	0.26	0.796	-.5098801	.6617301
27				.0083988	.2311591	0.04	0.971	-.4558979	.4726955
28				-.2127531	.1539994	-1.38	0.173	-.52207	.0965638
29				.2222519	.1765713	1.26	0.214	-.1324019	.5769057
30				-.696597	.2533265	-2.75	0.008	-1.205418	-.1877757
31				-.1856806	.3453238	-0.54	0.593	-.8792839	.5079227
32				.1671913	.6515857	0.26	0.799	-1.141557	1.47594
33				-2.971374	.6302908	-4.71	0.000	-4.237351	-1.705398
34				-1.081517	.4220763	-2.56	0.013	-1.929282	-.2337519
35				-.8538008	.1782373	-4.79	0.000	-1.211801	-.4958008
36				-2.11072	.2153852	-9.80	0.000	-2.543334	-1.678106
37				-.0724538	.2182202	-0.33	0.741	-.510762	.3658543
38				-2.087642	.2894568	-7.21	0.000	-2.669033	-1.506251
39				-.2748486	.2266876	-1.21	0.231	-.730164	.1804667
40				-.0365997	.1316638	-0.28	0.782	-.3010542	.2278548
41				-.8204083	.2665903	-3.08	0.003	-1.355871	-.2849459
42				.0952371	.2729022	0.35	0.729	-.4529031	.6433774
44				-.3479632	.3305076	-1.05	0.297	-1.011807	.3158808

45		-1.03916	.1748983	-5.94	0.000	-1.390454	-.6878669
46		-1.51665	.2116141	-7.17	0.000	-1.941689	-1.09161
47		.3278838	.0792663	4.14	0.000	.1686728	.4870948
48		-.2487844	.1929235	-1.29	0.203	-.6362826	.1387138
49		-.6264741	.2875367	-2.18	0.034	-1.204009	-.0489396
50		-1.201195	.3867603	-3.11	0.003	-1.978026	-.4243645
51		-.6199041	.3134695	-1.98	0.054	-1.249526	.0097178
53		0	(omitted)				
54		-1.148676	.1493309	-7.69	0.000	-1.448616	-.8487362
55		.5668324	.3933499	1.44	0.156	-.2232341	1.356899
56		-1.249674	.3478953	-3.59	0.001	-1.948442	-.5509054
t		0	(omitted)				
fip#c.t							
2		.0244529	.0230627	1.06	0.294	-.0218699	.0707757
4		-.030395	.0189648	-1.60	0.115	-.068487	.007697
5		.0128733	.004243	3.03	0.004	.0043509	.0213957
6		.007093	.0087707	0.81	0.423	-.0105234	.0247094
8		-.0071186	.0224994	-0.32	0.753	-.05231	.0380728
9		.0161712	.0453102	0.36	0.723	-.074837	.1071795
10		.0117182	.0197663	0.59	0.556	-.0279836	.05142
11		-.0032102	.0596737	-0.05	0.957	-.1230683	.1166479
12		-.0304593	.0165437	-1.84	0.072	-.0636884	.0027697
13		-.0388817	.014716	-2.64	0.011	-.0684396	-.0093238
15		.1492851	.013845	10.78	0.000	.1214766	.1770936
16		-.0746169	.01237	-6.03	0.000	-.0994627	-.0497711
17		.0345409	.0186901	1.85	0.071	-.0029992	.072081
18		.0100973	.0111422	0.91	0.369	-.0122824	.032477
19		.0149253	.0077716	1.92	0.061	-.0006845	.030535
20		.0292349	.0134707	2.17	0.035	.0021782	.0562916
21		-.0086793	.0068655	-1.26	0.212	-.0224691	.0051105
22		.0498841	.0101856	4.90	0.000	.0294257	.0703425
23		.0056534	.0214149	0.26	0.793	-.0373596	.0486664
24		.0440399	.0242775	1.81	0.076	-.004723	.0928028
25		-.0418595	.0361971	-1.16	0.253	-.1145636	.0308446
26		-.0361967	.0166168	-2.18	0.034	-.0695725	-.002821
27		.0162542	.0197497	0.82	0.414	-.0234143	.0559228
28		.0213194	.0078145	2.73	0.009	.0056236	.0370153
29		.0085451	.0086156	0.99	0.326	-.0087599	.0258501
30		-.0169654	.0105426	-1.61	0.114	-.0381409	.0042102
31		.0353449	.0193409	1.83	0.074	-.0035023	.0741922
32		-.05923	.0296812	-2.00	0.051	-.1188464	.0003864
33		.113456	.0268937	4.22	0.000	.0594385	.1674736
34		-.0147	.0362213	-0.41	0.687	-.0874527	.0580527
35		.005181	.0188962	0.27	0.785	-.0327731	.043135

36		.1242363	.0100497	12.36	0.000	.104051	.1444216
37		.0167285	.0174907	0.96	0.343	-.0184026	.0518595
38		.0392796	.0166212	2.36	0.022	.005895	.0726642
39		.0129426	.0154794	0.84	0.407	-.0181486	.0440338
40		.02788	.0106636	2.61	0.012	.0064614	.0492985
41		.0285276	.0195652	1.46	0.151	-.0107703	.0678255
42		-.018509	.0212777	-0.87	0.389	-.0612465	.0242284
44		.0009555	.0244091	0.04	0.969	-.0480715	.0499825
45		.0438337	.008594	5.10	0.000	.0265721	.0610952
46		.0462614	.0174768	2.65	0.011	.0111581	.0813646
47		-.0210074	.0091351	-2.30	0.026	-.0393557	-.002659
48		.007767	.0158324	0.49	0.626	-.0240332	.0395673
49		-.0864221	.0160945	-5.37	0.000	-.1187488	-.0540954
50		.0000726	.0221555	0.00	0.997	-.0444281	.0445733
51		.0198732	.019582	1.01	0.315	-.0194583	.0592048
53		0	(omitted)				
54		.0671996	.0113235	5.93	0.000	.0444556	.0899436
55		-.0154737	.0200596	-0.77	0.444	-.0557645	.0248172
56		.0073057	.0281259	0.26	0.796	-.0491867	.0637982
acc		-.0011368	.0020799	-0.55	0.587	-.0053145	.0030409
pi		-.0244522	.0468311	-0.52	0.604	-.1185152	.0696109
ir		-.0000504	.000101	-0.50	0.620	-.0002533	.0001525
alcohol		-.024153	.1789552	-0.13	0.893	-.383595	.335289
crack		.0418238	.0366812	1.14	0.260	-.0318526	.1155002
poverty		.0181194	.0280102	0.65	0.521	-.0381408	.0743797
income		.0000232	.0000526	0.44	0.661	-.0000824	.0001289
ur		-.0428437	.0378736	-1.13	0.263	-.118915	.0332277
_cons		7.994616	.959291	8.33	0.000	6.067823	9.921408

We reject the null hypothesis that coefficients for the DDD estimators in 1986-1989 are 0 for 15-19 year olds, and we conclude that the rates of long-term gonorrhea declines in the earlier adopters at the 5% level for 15-19 year olds.

We do notice that we reject the null hypothesis for parameters in 1991-1992 for 25-29 year olds for `i.repeal##i.year` interactions. This can possibly be a concern.

Next, we will plot our DDD parameters estimates.

```
xi: reg lnr i.repeal*i.year i.younger*i.repeal i.younger*i.year i.yr*i.year ///
i.fip*t acc pi ir alcohol crack poverty income ur if bf==1 & (age==15 | age==25) ///
[aweight=totpop], cluster(fip)
```

```
*Parameter Estimate into dataframe
```

```

parwest, label for(estimate min95 max95 %8.2f) li(parm label estimate min95 max95) sav:

*Get Parameter Estimate Dataframe
use ./bf15_DDD.dta, replace

*Keep Triple Difference-in-Differences Parameter Estimates only
keep in 82/96

gen      year=.
replace year=1986 in 1
replace year=1987 in 2
replace year=1988 in 3
replace year=1989 in 4
replace year=1990 in 5
replace year=1991 in 6
replace year=1992 in 7
replace year=1993 in 8
replace year=1994 in 9
replace year=1995 in 10
replace year=1996 in 11
replace year=1997 in 12
replace year=1998 in 13
replace year=1999 in 14
replace year=2000 in 15

sort year

*Graph the Triple Diff-in-Diff
twoway (scatter estimate year, mlabel(year) mlabsize(vsmall) msize(tiny)) ///
      (rcap min95 max95 year, msize(vsmall)), ///
      ytitle(Repeal x 20-24yo x year estimated coefficient) yscale(titlegap(2)) ///
      yline(0, lwidth(thin) lcolor(black)) xtitle(Year) ///
      xline(1986 1987 1988 1989 1990 1991 1992, lwidth(vvvthick) lpattern(solid) ///
      lcolor(ltblue)) xscale(titlegap(2)) ///
      title(Estimated effect of abortion legalization on gonorrhea) ///
      subtitle(Black females 15-19 year-olds) ///
      note(Whisker plots are estimated coefficients of DDD estimator from Column b of Table
      legend(off)

```

It is interesting that we fail to reject the null hypothesis for 15-19 year olds in 1991-1992, but we reject the null hypothesis for 25-29 year olds in 1991-1992.

We can run the estimates for White women, Black males, and White males. The code is the same as above with the exception of the subset.

We fail to reject the null hypothesis that parameters for the DDD estimator are

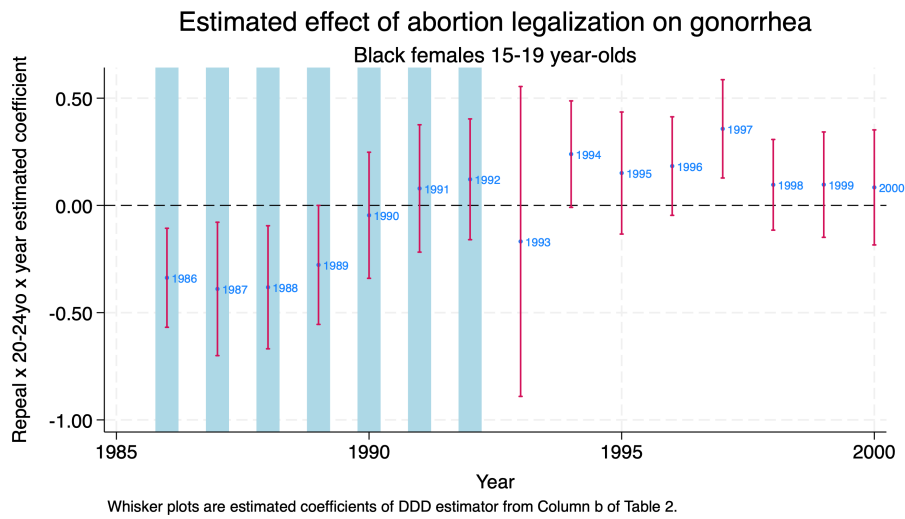


Figure 1.1: Parameter estimates for the DDD estimator for Black Females

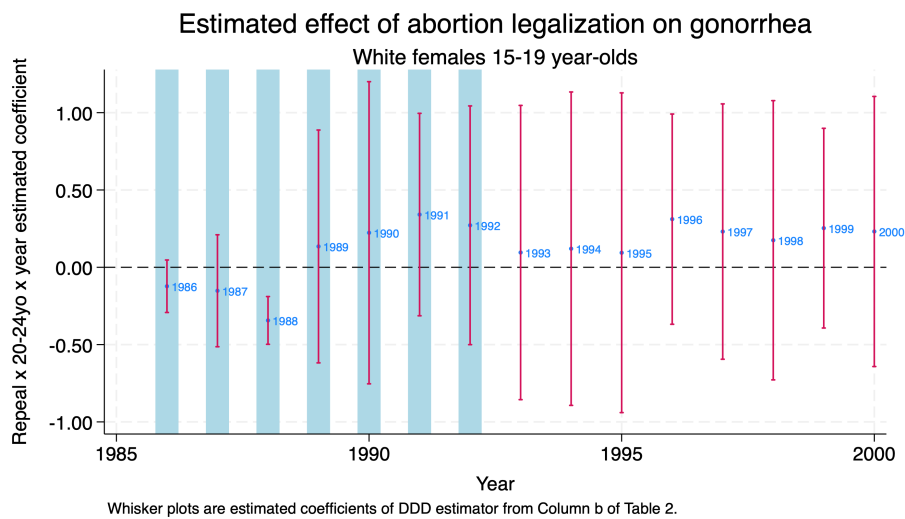


Figure 1.2: Parameter estimates for White Women

0 with the exception of 1988.

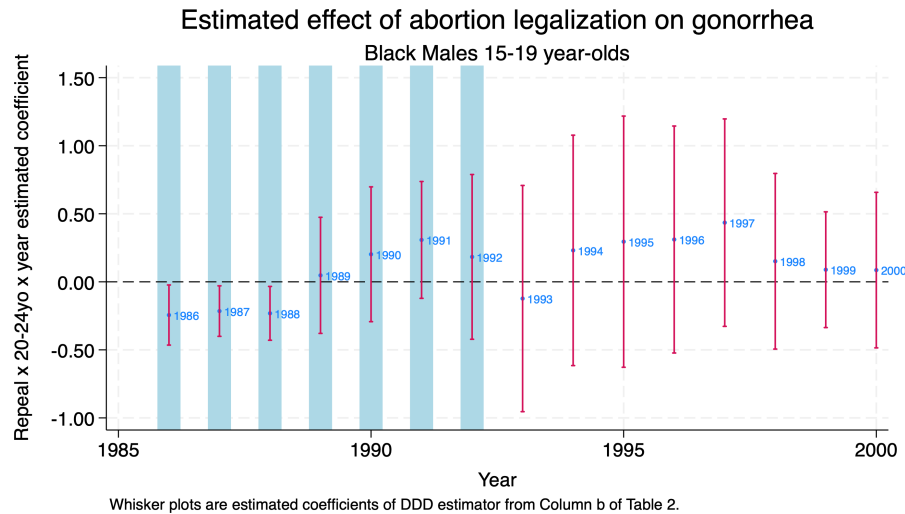


Figure 1.3: Parameter estimates for Black Males

We reject the null hypothesis that the parameters of the DDD estimator is 0 for 1986-1988, but we fail to reject the null hypothesis for years 1989-1992.

We fail to reject the null hypothesis that the parameters of the DDD estimator are 0 for White Males, except for 1986.

These results show that the policy impacts may vary by demographics. However, these tests show some support to what Cunningham and Cornwell (2013) hypothesis. However, as Cunningham (2021) notes the abortion hypothesis makes very specific predictions, and we fail to find a parabolic relationship.

1.3 20-24 year olds vs 25-29 year olds

Next, we will conduct the same exercise, but we will focus on 20-24 year olds.

```
quietly {
use "/Users/Sam/Desktop/Econ 672/Data/abortion.dta", clear
```

```
Second DDD model for 20-24 year olds vs 25-29 year olds black females in repeal vs Roe
gen younger2 = 0
replace younger2 = 1 if age == 20
gen yr2=(repeal==1) & (younger2==1)
```

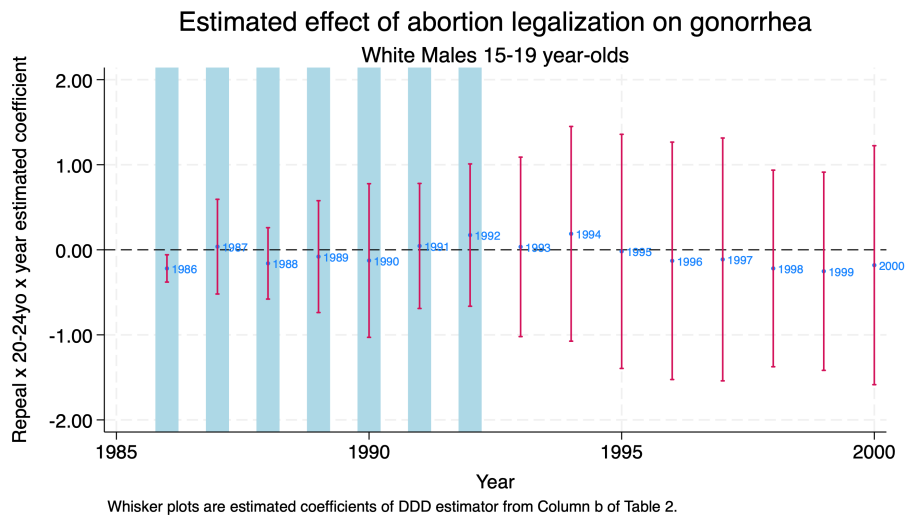



Figure 1.4: Parameter Estimates for White Males

```

gen wm=(wht==1) & (male==1)
gen wf=(wht==1) & (male==0)
gen bm=(wht==0) & (male==1)
gen bf=(wht==0) & (male==0)
char year[omit] 1985
char repeal[omit] 0
char younger2[omit] 0
char fip[omit] 1
char fa[omit] 0
char yr2[omit] 0

*Triple DDD
*Compare 20-24 to 25-29 year olds in repeal vs Roe states
*OR
reg lnr i.repeal##i.year##i.younger2 i.fip##c.t acc pi ir alcohol crack poverty ///
income ur if bf==1 & (age==20 | age==25) [aweight=totpop], cluster(fip)
*XI
xi: reg lnr i.repeal*i.year i.younger2*i.repeal i.younger2*i.year i.yr2*i.year ///
i.fip*t acc pi ir alcohol crack poverty income ur if bf==1 & (age==20 | age==25) ///
[aweight=totpop], cluster(fip)

parmest, label for(estimate min95 max95 %8.2f) li(parm label estimate min95 max95) saving(bf20_D)

*Get Parameter Estimate Dataframe

```

```

use ./bf20_DDD.dta, replace

keep in 82/96

gen      year=.
replace year=1986 in 1
replace year=1987 in 2
replace year=1988 in 3
replace year=1989 in 4
replace year=1990 in 5
replace year=1991 in 6
replace year=1992 in 7
replace year=1993 in 8
replace year=1994 in 9
replace year=1995 in 10
replace year=1996 in 11
replace year=1997 in 12
replace year=1998 in 13
replace year=1999 in 14
replace year=2000 in 15

sort year
}
*Graph the Triple Diff-in-Diff
twoway (scatter estimate year, mlabel(year) mlabsize(vsmall) msize(tiny)) ///
      (rcap min95 max95 year, msize(vsmall)), ///
      ytitle(Repeal x 20-24yo x year estimated coefficient) yscale(titlegap(2)) ///
      yline(0, lwidth(thin) lcolor(black)) xtitle(Year) ///
      xline(1991 1992 1993 1994 1995 1996 1997, lwidth(vvvthick) lpattern(solid) ///
      lcolor(ltblue)) xscale(titlegap(2)) ///
      title(Estimated effect of abortion legalization on gonorrhea) ///
      subtitle(Black Females 20-24 year-olds) ///
      note(Whisker plots are estimated coefficients of DDD estimator from Column b of Table
      legend(off)

```

We fail to reject the null hypothesis in the time period of interest. One concern is that 20-24 year olds are too close in age to 25-29 year olds for a comparison. If these groups are interacting, then we will have a spillover effect. This spillover effect will violate our Stable Unit Treatment Value Assumptions (SUTVA).

The abortion hypothesis provides very specific predictions, and our results show some skepticism for the abortion hypothesis explaining long-term gonorrhea incidences.

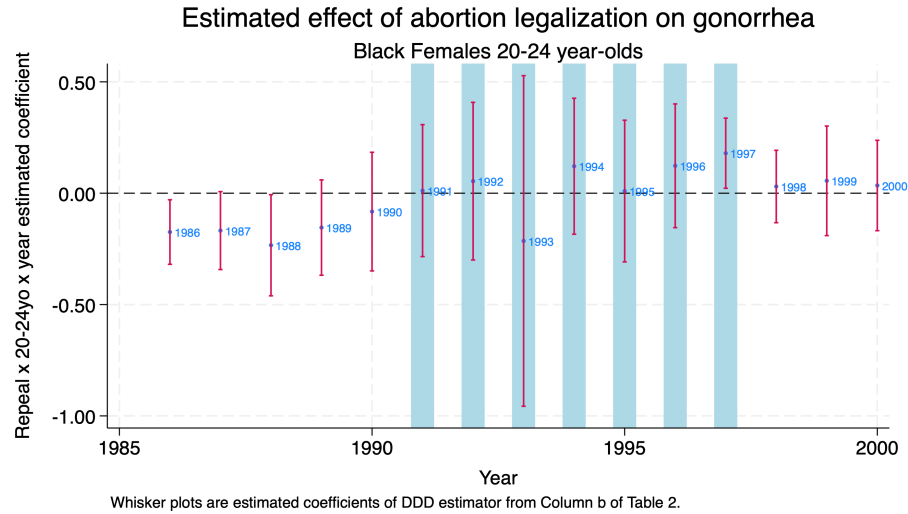


Figure 1.5: Parameter estimates for the DDD estimator for 20-24 Black Females

Chapter 2

Event Studies

We can use panel event studies to assess pre-treatment trends. (Please do not confuse panel event studies with time series event studies). We will use data from the National Longitudinal Survey for Women to assess the impact of unionization on wages.

Luckily, we have some Stata commands that will help us with event studies. I have found these commands easier to use than the syntax provided by Cunningham (2021). Freyaldenhoven, Hansen, Perez, and Shapiro (2021) provide their command package *xtevent* to estimate the parameters of an event study, plot the estimates, and test the estimates.

```
ssc install xtevent
```

2.1 National Longitudinal Survey

First, we will need to import the data and inspect the dataset.

```
cd "/Users/Sam/Desktop/Econ 672/Data/"
use nlswork
describe
```

```
/Users/Sam/Desktop/Econ 672/Data
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)
```

```
Contains data from nlswork.dta
```

```

Observations:      28,534      National Longitudinal Survey of
                               Young Women, 14-24 years old in
                               1968
Variables:         21         13 Apr 2026 19:21
                               (_dta has notes)

```

Variable name	Storage type	Display format	Value label	Variable label
idcode	int	%8.0g		NLS ID
year	byte	%8.0g		Interview year
birth_yr	byte	%8.0g		Birth year
age	byte	%8.0g		Age in current year
race	byte	%8.0g	racelbl	Race
msp	byte	%8.0g		1 if married, spouse present
nev_mar	byte	%8.0g		1 if never married
grade	byte	%8.0g		Current grade completed
collgrad	byte	%8.0g		1 if college graduate
not_smsa	byte	%8.0g		1 if not SMSA
c_city	byte	%8.0g		1 if central city
south	byte	%8.0g		1 if south
ind_code	byte	%8.0g		Industry of employment
occ_code	byte	%8.0g		Occupation
union	byte	%8.0g		1 if union
wks_ue	byte	%8.0g		Weeks unemployed last year
ttl_exp	float	%9.0g		Total work experience
tenure	float	%9.0g		Job tenure, in years
hours	int	%8.0g		Usual hours worked
wks_work	int	%8.0g		Weeks worked last year
ln_wage	float	%9.0g		ln(wage/GNP deflator)

Sorted by: idcode year

We can see that this is a longitudinal survey of women ages 14-24 in 1964. Let's inspect the outcome of interest, the natural log of wages.

```

sum ln_wages, detail
histogram ln_wage, title(Natural Log of Wages) caption(NLS Women Ages 14-24 1964)

```

```

variable ln_wages not found
r(111);

r(111);

```

Next, we will set up the panel

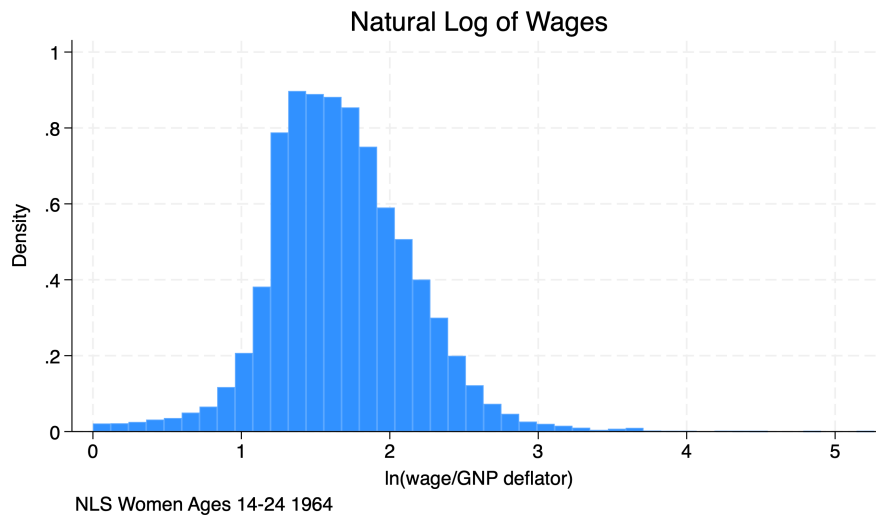


Figure 2.1: Histogram of Natural Log of Wages

```
by idcode (year): gen time=_n
xtset idcode time
```

```
Panel variable: idcode (unbalanced)
Time variable: time, 1 to 15
Delta: 1 unit
```

Generate a policy variable that follows staggered adoption. This observe changes in the union status or treatment variable of interest.

```
by idcode (time): gen union2 = sum(union)
replace union2 = 1 if union2 > 1
tab year union2
```

(4,177 real changes made)

Interview	union2		
year	0	1	Total
68	1,375	0	1,375
69	1,232	0	1,232
70	1,508	178	1,686

71		1,559	292		1,851
72		1,345	348		1,693
73		1,561	420		1,981
75		1,787	354		2,141
77		1,595	576		2,171
78		1,390	574		1,964
80		1,124	723		1,847
82		1,245	840		2,085
83		1,171	816		1,987
85		1,184	901		2,085
87		1,213	951		2,164
88		1,270	1,002		2,272
-----+-----+-----					
Total		20,559	7,975		28,534

2.2 *xtevent*

We can utilize the command *xtevent* to calculate and plot an panel event study instead of complex syntax. The command is the following:

```
xtevent depvar [indepvars], policyvar(PostTreatVar) panelvar(idvar)
timevar(timevarname)
```

We have several options that we need to consider as well. One of the key option is the *window* options. We will set the window to 3 and then rerun the *xtevent* command with a window of 10.

```
xtevent ln_w age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure , pol(union2) w(3) clu
```

Using options panelvar and timevar from xtset

No proxy or instruments provided. Implementing OLS estimator

Linear regression, absorbing indicators	Number of obs	= 23,541
Absorbed variable: idcode	No. of categories	= 2,771
	F(27, 2770)	= 68.03
	Prob > F	= 0.0000
	R-squared	= 0.6662
	Adj R-squared	= 0.6212
	Root MSE	= 0.2872

(Std. err. adjusted for 2,771 clusters in idcode)

		Robust			
ln_wage		Coefficient	std. err.	t	P> t
					[95% conf. interval]

_k_eq_m4		-.0449924	.0184471	-2.44	0.015	-.0811639	-.0088209
_k_eq_m3		-.0565855	.0168862	-3.35	0.001	-.0896964	-.0234746
_k_eq_m2		-.0421016	.0134889	-3.12	0.002	-.0685508	-.0156524
_k_eq_p0		.089023	.0116097	7.67	0.000	.0662585	.1117874
_k_eq_p1		.0815659	.0138364	5.90	0.000	.0544352	.1086966
_k_eq_p2		.0804259	.0153595	5.24	0.000	.0503086	.1105431
_k_eq_p3		.0635638	.0172566	3.68	0.000	.0297266	.097401
_k_eq_p4		.0299474	.0202315	1.48	0.139	-.0097229	.0696178
age		.0200353	.0067278	2.98	0.003	.0068434	.0332273
c.age#c.age		-.0005226	.0001023	-5.11	0.000	-.0007231	-.000322
t1l_exp		.0491603	.0054054	9.09	0.000	.0385614	.0597593
c.t1l_exp#							
c.t1l_exp		-.0004479	.000193	-2.32	0.020	-.0008264	-.0000695
tenure		.0095845	.0015872	6.04	0.000	.0064723	.0126967
time							
2		.0560166	.0088315	6.34	0.000	.0386996	.0733335
3		.0575122	.0129664	4.44	0.000	.0320874	.082937
4		.0603109	.0171093	3.53	0.000	.0267627	.0938592
5		.0485817	.0215347	2.26	0.024	.0063561	.0908074
6		.0484417	.0263678	1.84	0.066	-.0032608	.1001441
7		.0293154	.03143	0.93	0.351	-.0323131	.0909439
8		.021307	.0362073	0.59	0.556	-.0496891	.0923031
9		.0151769	.0409211	0.37	0.711	-.0650621	.0954159
10		-.0014045	.045488	-0.03	0.975	-.0905984	.0877894
11		-.0124446	.0507805	-0.25	0.806	-.112016	.0871269
12		-.0173009	.0561008	-0.31	0.758	-.1273046	.0927028
13		-.0346925	.0608558	-0.57	0.569	-.1540199	.0846348
14		-.0097838	.0668818	-0.15	0.884	-.140927	.1213595
15		-.0288719	.0799116	-0.36	0.718	-.1855643	.1278204
_cons		1.221736	.1016406	12.02	0.000	1.022437	1.421035

From the event study, we find that unionization increases wages for women by $(e^{0.089} - 1) * 100\% = 9.8\%$ to about $(e^{0.06356} - 1) * 100\% = 6.6\%$ for four years after unionization.

Now, we will expand the window to 10 years.

```
xtevent ln_w age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure , pol(union2) w(10) cl
```

Using options panelvar and timevar from xtset

No proxy or instruments provided. Implementing OLS estimator

Linear regression, absorbing indicators	Number of obs	=	6,687
Absorbed variable: idcode	No. of categories	=	510
	F(41, 509)	=	18.19
	Prob > F	=	0.0000
	R-squared	=	0.6709
	Adj R-squared	=	0.6414
	Root MSE	=	0.2626

(Std. err. adjusted for 510 clusters in idcode)

	ln_wage	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
_k_eq_m11		-.0935686	.0915262	-1.02	0.307	-.2733842	.086247
_k_eq_m10		-.0700651	.0749131	-0.94	0.350	-.217242	.0771118
_k_eq_m9		-.052819	.0531611	-0.99	0.321	-.1572612	.0516232
_k_eq_m8		-.0481707	.0461557	-1.04	0.297	-.1388498	.0425083
_k_eq_m7		-.0709862	.0420374	-1.69	0.092	-.1535744	.0116021
_k_eq_m6		-.0815994	.0394746	-2.07	0.039	-.1591527	-.0040462
_k_eq_m5		-.0735276	.0330956	-2.22	0.027	-.1385483	-.0085069
_k_eq_m4		-.0792023	.0321908	-2.46	0.014	-.1424454	-.0159591
_k_eq_m3		-.066719	.0278508	-2.40	0.017	-.1214358	-.0120023
_k_eq_m2		-.0458597	.0243467	-1.88	0.060	-.0936922	.0019727
_k_eq_p0		.0820767	.0208546	3.94	0.000	.041105	.1230485
_k_eq_p1		.0836871	.0229813	3.64	0.000	.0385372	.1288369
_k_eq_p2		.0760776	.0248914	3.06	0.002	.027175	.1249802
_k_eq_p3		.0453932	.027055	1.68	0.094	-.0077599	.0985464
_k_eq_p4		.0314656	.0335141	0.94	0.348	-.0343774	.0973086
_k_eq_p5		.0592412	.0330584	1.79	0.074	-.0057065	.1241889
_k_eq_p6		.0298114	.0366003	0.81	0.416	-.0420949	.1017177
_k_eq_p7		.0346491	.0409492	0.85	0.398	-.0458012	.1150995
_k_eq_p8		.0318303	.0456673	0.70	0.486	-.0578893	.1215499
_k_eq_p9		.0263062	.0526183	0.50	0.617	-.0770695	.129682
_k_eq_p10		.0151391	.0553342	0.27	0.785	-.0935724	.1238507
_k_eq_p11		.0007066	.0605004	0.01	0.991	-.1181547	.119568
age		.0287481	.0175581	1.64	0.102	-.0057472	.0632434
c.age#c.age		-.0008166	.0002452	-3.33	0.001	-.0012984	-.0003349

t1l_exp		.0506316	.0107874	4.69	0.000	.0294383	.0718249
c.t1l_exp#							
c.t1l_exp		-.0003699	.0003314	-1.12	0.265	-.001021	.0002812
tenure		.0062955	.002426	2.60	0.010	.0015294	.0110616
time							
2		.0701101	.018315	3.83	0.000	.0341277	.1060924
3		.0573137	.0288765	1.98	0.048	.0005819	.1140455
4		.0617034	.039705	1.55	0.121	-.0163024	.1397092
5		.0600686	.0524503	1.15	0.253	-.0429772	.1631144
6		.0695395	.0659455	1.05	0.292	-.0600194	.1990985
7		.0525483	.0821128	0.64	0.522	-.1087734	.21387
8		.0604927	.0978685	0.62	0.537	-.1317833	.2527686
9		.0688636	.1110785	0.62	0.536	-.1493652	.2870925
10		.0520036	.127703	0.41	0.684	-.1988862	.3028935
11		.0646191	.144279	0.45	0.654	-.2188365	.3480747
12		.0959642	.1585038	0.61	0.545	-.215438	.4073664
13		.1001231	.1730002	0.58	0.563	-.2397593	.4400056
14		.1470231	.1898777	0.77	0.439	-.2260174	.5200636
15		.1433274	.2071761	0.69	0.489	-.2636981	.5503529
_cons		1.223692	.2833243	4.32	0.000	.6670628	1.780321

We can see the range of outcomes is similar, but for only three years after the event.

2.3 *xteventplot*

We can plot the results from the *xtevent* command in two ways. One way is to add the option *plot* to the *xtevent* command. The second way is to use the command *xteventplot*, which I prefer due to more options.

We can visualize the pretreatment trends to see if parallel trends holds.

```
quietly xtevent ln_w age c.age#c.age t1l_exp c.t1l_exp#c.t1l_exp tenure , pol(union2) w(3) cluster(union2)
xteventplot, title("Event Study of Unionization on Wages") caption("NLS Women 14-24 in 1964")
```

From our plot, we can easily see that we reject the null hypothesis that our pretreatment parameters are equal to 0.

We will run it with a window of 10 years for pretreatment.

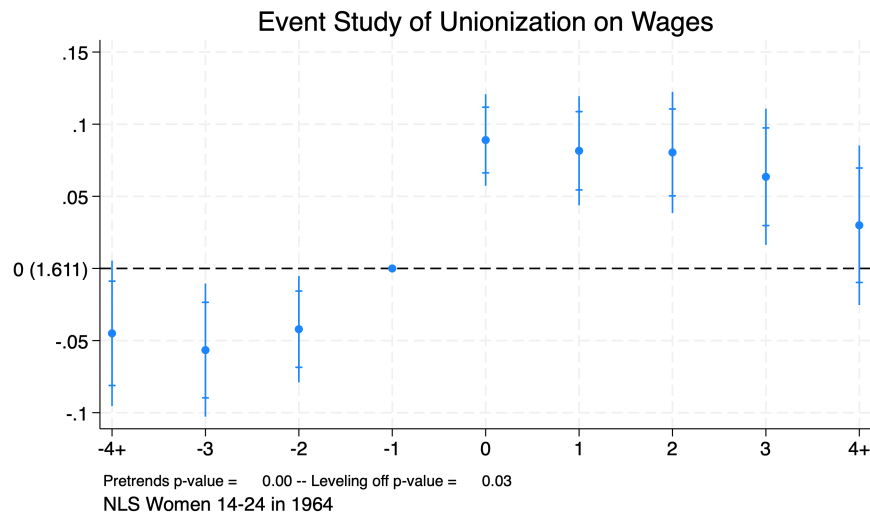


Figure 2.2: Event Study with 3 years pretreatment

```
xtevent ln_w age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure , pol(union2) w(10) cl
xteventplot, title("Event Study of Unionization on Wages") caption("NLS Women 14-24 in
```

Our plot shows that we reject the null hypothesis for the pretreatment years 3-5. We need to be concerned that the parallel trends assumption does not hold.

2.4 *xteventtest*

We can test the null hypothesis that the cumulative pretreatment variables are equal to 0 with the command *xteventtest* command. We can add the options *allpre* and *allpre cumul* to test our pretreatment trends.

These conduct *F*-tests to test the joint exclusion restriction, which is similar to what we did in Econ 645.

We'll use the 3-year pretreatment window.

```
quietly xtevent ln_w age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure , pol(union2) w
xteventtest, allpre cumul
```

Test for all pre-event coefficients = 0

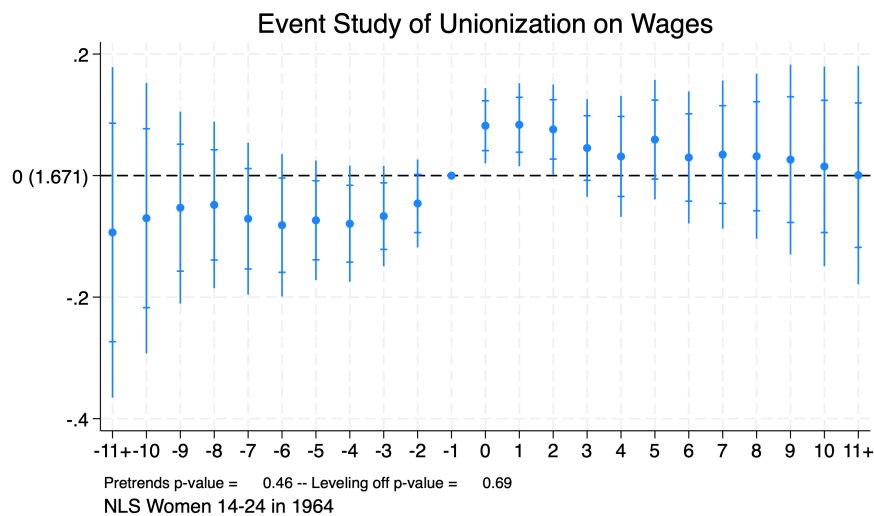


Figure 2.3: Event Study with 10 years pretreatment

Test sums of coefficients

```
( 1)  _k_eq_m4 + _k_eq_m3 + _k_eq_m2 = 0
```

```
F( 1, 2770) = 12.11
Prob > F = 0.0005
```

Our results from the F -test show that we reject the null hypothesis that the cumulative pretreatment trend is 0.

We'll test this for the 10-year pretreatment window.

```
quietly xtevent ln_w age c.age#c.age ttl_exp c.ttl_exp#c.ttl_exp tenure , pol(union2) w(10) clust
xteventtest, allpre cumul
```

Test for all pre-event coefficients = 0

Test sums of coefficients

```
( 1)  _k_eq_m11 + _k_eq_m10 + _k_eq_m9 + _k_eq_m8 + _k_eq_m7 + _k_eq_m6 +
      _k_eq_m5 + _k_eq_m4 + _k_eq_m3 + _k_eq_m2 = 0
```

```
F( 1, 509) = 4.42
Prob > F = 0.0360
```

Again, our results from the F -test show that we reject the null hypothesis that the cumulative pretreatment trend is 0.

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Freyaldenhoven, S., Hansen C. & Perez, JEP., and Shapiro J. (2021). *XTEVENT: Stata module to estimate and visualize linear panel event-study models*. Statistical Software Components S458987, Boston College Department of Economics, revised 12 Jul 2024. <https://ideas.repec.org/c/boc/bocode/s458987.html>

Gruber, Levine, and Staiger (1999). Gruber, J., Levine, P.B., and Staiger, D. (1999). Abortion Legalization and Child Living Circumstances: Who Is the ‘Marginal Child’? *The Quarterly Journal of Economics* 114 (1): 263–91.

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